PHILIP MIROWSKI

Postmodernism and the social theory of value

While the other contributors to this symposium have focused upon issues of epistemology, canon creation in the history of economic thought, and the metaphors governing our cultural images of ownership and control, I should like to propose that the implications of the cultural changes that travel under the rubric of "postmodernism" could potentially transcend a critique of orthodox economics, to the point of initiating an alternative to the neoclassical theory of value. If it is a hallmark of postmodernism to deny the text has a single fixed and stable referent, then what if the "economy" were treated in a similar manner? And, indeed, what if this reconceptualization went beyond discussions of methodology to actually frame an alternative mathematical formalization of the notion of value?

I have argued elsewhere (Mirowski, 1989, 1990a) that, far from there being a surfeit of theories of value to choose from in the history of economic thought, from the quantitative/analytic point of view there really have only been two: the "substance theory of value," exemplified by such particular instances as the physiocratic corn theory and the classical labor-embodied theory of value; and the "field theory of value," best exemplified by the neoclassical theory of utility. There has also been a third subterranean (and relatively ineffectual) current that denies the existence of the phenomenon of value altogether, exemplified

The author is Carl Koch Professor of Economics and the History and Philosophy of Science at the University of Notre Dame, Notre Dame, Indiana. He would like to thank David Ellerman for helpful comments and discussion, without implicating him in any of the author's own errors; he is also not responsible for any uses to which his own work has been put in this paper. Various audiences have played a part in shaping this outcome, including the Seminar in Institutionalist Economics at the University of Tennessee and the auditors of Economics 785b at Yale University. The literary "dual" to the present mathematical argument may be found in Mirowski, 1990a.

by the writings of Samuel Bailey and William Thornton. The pitfalls and attractions of this radical abjuration of "value" have also been discussed elsewhere (Mirowski, 1991).

The primary reason for such slim pickings can be traced to the Durkheim–Mauss–Douglas thesis (Mirowski, 1988, p. 109), which predicts that the social categories employed to organize our discourse would be reflections of the categories we use to discuss the natural world. The structure of the classical substance and of the neoclassical field theories were both largely projections of mathematical models of the physical world dominant in those respective eras. In other words, the social structures of the economy were invested with the determinacy and lawlike character of things. I would claim that (with very few partial exceptions, such as the American institutionalists) there has *never* been a serious exploration of the logical structure of a thoroughgoing social theory of value. It would be explicitly "social," and perhaps even "postmodern," because it would refrain from grounding any aspect of value either in the "natural" attributes of the commodities (the substance theories), or in the supposed inherent psychological regularities of the individual mind (neoclassical field theory). Instead, it would opt for the third modality of rooting the structure of value in contingent social institutions. As Mary Douglas (1986) has explained, institutions are founded upon analogy and provide the conceptual identity for phenomena. While Douglas most likely did not intend the technical mathematical definition of "identity" (i.e., one of the requirements in abstract algebra for the existence of a group), I do.

I. Four central issues in value theory

The full justification of the quantification of prices is profoundly more complicated than the rather cursory discussion found in Mirowski (1986). There seem to be four somewhat separable issues, each demanding the use of a different mathematical formalism to be expressed properly. The first is the problem of the constitution of the identity of the commodity, which would be the province of *measurement theory*. The second is the problem of conceptualizing trade as an ongoing network of permitted and blocked exchanges of idealized classes of goods between individual human beings. This question would be the province of *graph theory*. The third is the description of the institution of "value" as distinct from, yet predicated upon, the previous two issues. This would largely be a question of *abstract algebra*. The social theory of value would be consolidated by a concatenation of the above devices. Far from being a pipe dream, the way forward has already been indicated by Ellerman (1984, 1988) using the formalism of *directed graphs* under the rubric of "arbitrage theory." Fourth, the irreducibly contingent character of value would be expressed by linking the above formalisms to the structures of *probability theory*, by revealing that probability is itself a special case of arbitrage theory.

It is important at this point to be self-consciously skeptical about this or any other program that imports still more relatively unfamiliar mathematical devices into a situation where the profession is already rife with superfluous and ill-considered abstract mathematical formalisms (Mirowski, 1986, 1991). In orthodox neoclassical economics, there is no serious intellectual justification for this escalation of technique, other than to disguise further its origins in nineteenth-century physics, or to provide a pecking order within the discipline and perhaps to exclude the curious layman. Here the role of mathematics will be different. The formalisms sketched out in these notes are required because it is a historical fact that modern market actors predicate their economic interactions upon prices and quantities expressed as rational numbers. This central fact of markets must have an explanation grounded in the theory of social institutions. But more significantly, mathematics is a superior method of the translation of metaphors between disciplines (Mirowski, 1988, ch. 8); and a major problem of a social theory of value is to reorient the culture away from its failed naturalism. Hence, it will be no accident that most of the formalisms proposed here are *not* generally found in physics.

II. The identity of the commodity

This first problem has almost never been dispassionately entertained in the economics literature, since it has simply been assumed that the categorical identity of any commodity was immediately transparently obvious to everyone and determined by external natural attributes. However, as argued in Mirowski (1986), not all apples are alike; not even all Macintoshes are identical. The very idea that identity is unproblematically determinate violates the purported methodological individualism of neoclassical theory: has anyone ever asked you if you thought all bottled water was "alike"? The question then arises: why would a market system be driven to standardize commodities and make them interchangeable in the social sphere (if not for each individual)?

The way commodities have been measured for purposes of trade has a very strange history (Kula, 1986). Briefly, commodity identities seem to have evolved from definitions tied to individual personalities (e.g., the feudal lord defined a bushel of grain differently for each person), to more relatively impersonal measures, culminating in the metric system in the French Revolution. This history can be seen as one chapter in a progressive attempt socially to constitute the identity of the commodity, and more precisely to remove it from the sphere of the personal and idiosyncratic. Such abstraction from the realm of the personal is an absolute prerequisite to rendering market trade quantitative.

There is no one "correct" way for a society to measure a commodity, and the way its measurement is instituted has important consequences for its subsequent manipulation in various formal mathematical schemes. It may be represented on an interval, ordinal, ratio, or absolute scale (Roberts, 1979); the attribute chosen to ground the measurement may be continuous or discrete; it may be reversible or irreversible. All of these considerations are the subject of a mathematical discipline called *measurement theory* (Roberts, 1979; Osborne, 1978). Measurement theory has appeared in a fringe literature in economics (de Jong, 1967; Luce and Narens, 1987), but almost exclusively in the context of trying to characterize utility; and therefore this literature has seemed somewhat superfluous. Instead, our proposed social theory of value would use it to highlight the difficulty in getting a society to measure something consistently, and to establish that this standardization is an important aspect of the spread of the market, ranging from the imposition of a metric system to the persistent drive down to the present day to standardize commodity identities through mass production, advertising, and the elimination of the craft character of precapitalist production. On a more pragmatic note, measurement theory would also be used to check that equations in the social theory of value are always dimensionally conformable. (This would prevent such travesties as $Y = (K,L)$.) Finally, in a world where, because of their contingent historical character, measurement scales are arbitrarily mixed, such justifications of the formal basis of economics such as that of Gerard Debreu would be barred as simply fallacious:

> Having chosen a unit of measurement for each one of them [the commodities], and a sign convention to distinguish inputs from outputs, one can describe the action of an economic agent by a vector in the commod-

> ity space R^l. The fact that the commodity space has the structure of a real vector space is a basic reason for the success of the mathematicization of economic theory. [Debreu, 1984, pp. 267-268]

The point of departure of a social theory of value should be: commodities must be rendered socially "measurable," preferably on some interval scale, in order for trade to be quantified and value to be defined. Because of the heterogeneity of individual perceptions, of scales and of their conventions, commodities do not genuinely span a real vector space at any specific historical nexus. And, in any event, the notion of an independent natural metric in "commodity space" is meaningless. Hence, value cannot be grounded in or deduced from the nature of the commodity itself: it cannot be collapsed to a problem of arbitrarily picking a *numeraire*. Commodities are rendered quantitative in the marketplace as part of the construction of standardized production and marketing, or, in other words, as part and parcel of the fabrication of value.

III. The network of trade

There is a sporadic and underdeveloped literature on the "dual coincidence of wants" and the problems of getting the trading activities of numerous individual actors to mesh. Contrary to the claims of neoclassical theory to base itself on individual behavior and the coordination activities of a market, it has had little to say about this problem (Clower, 1984; Ostroy and Starr, 1974; Eckalbar, 1984). The reason for this neglect is that the enumeration of the permutations of the possible channels of trade is itself another issue that should be formalized prior to the inscription of value theory. The problem of who may deal with whom on a social basis is qualitatively different from the "effective demand" problem of who has the resources to engage in exchange.

In order to guarantee that we don't lose sight of the people in the economy, let us begin by defining a "social traffic" matrix $\mathbf{z}$ where the rows $i = 1, \ldots, M$ index the people in our market and the columns $j = 1, \ldots, N$ index the various classes of commodities that have been subjected to the local process of standardization and measurement described in the section above. With the following sign convention, the various entries in this matrix tell us who stands willing and able to trade each commodity. Let

$$z_{ij} = \begin{cases} 1 \text{ if person } i \text{ will sell good } j \text{ in trade;} \\ -1 \text{ if person } i \text{ will buy good } j \text{ in trade;} \\ 0 \text{ if person } i \text{ will not trade good } j. \end{cases}$$

It is well known in the specialist neoclassical literature that certain configurations of the **z** matrix—namely those where the dual coincidence of wants is frustrated—would stymie exchange, even in the presence of an "equilibrium" vector of Walrasian prices.[1] A simple example is provided by the following three-person, three-good matrix, $\mathbf{z}^*$:

$$\mathbf{z}^* = \begin{bmatrix} 1 & -1 & 0 \\ -1 & 0 & 1 \\ 0 & 1 & -1 \end{bmatrix}$$

Here, person 1 would like to sell good one and buy good two, but neither of the other participants can both take good one off his hands and give him something he wants in return, even in any pairwise sequence of exchanges. The existing literature has tried to get around this problem in one of three ways: (a) by simply presuming that everyone starts out with sufficient stocks of all commodities and/or effectively wants everything all the time; (b) by positing the existence of a trade coordinator(s) who stands ready to buy and sell unlimited amounts of a designated set of commodities; or (c) by positing the existence of a commodity that everyone stands ready to buy and sell at all times (making everyone a trade coordinator). I find none of these expedients historically plausible or especially theoretically interesting, especially given that the posited trade coordinator looks a bit too much like the inexplicably benevolent and other-worldly Walrasian auctioneer, and the universally acceptable commodity does not capture all the signifi-

[1] For instance, Eckalbar (1984), and Davidson (1978, ch. 6). Further, since Ostroy and Starr (1974) take prices as given exogenously and all equal to unity, abjuring all discussions of utility, their matrices all qualify as social traffic matrices; therefore, many of their results may be absorbed into the social theory of value with little or no alteration. For instance, they demonstrate that simple coordination requires either (a) the existence of a commodity that everyone is willing to buy and sell and keep appreciable stocks for trading purposes; or (b) the existence of a trade coordinator who is willing to buy or sell unlimited quantities of most goods. Since neither situation occurs "naturally" (or, as they phrase it, decentralization is impossible in a pure barter economy), the moral of their work is that one cannot take a "connected" trading system for granted; rather, institutions must be created in order to guarantee its existence.

cant functions of money. The social traffic matrix does keep us aware that money can reduce the number of pairwise trades necessary to arrive at any particular commodity; but it will also serve to demonstrate that all the problems of market coordination are not therein dissolved, as we shall witness below in section VI.

Another aspect of the social traffic matrix critical for the subject of section V below is the existence of "circuits" in the matrices. A "circuit" would describe the ability to arrange a sequence of trades in a circle in order to arrive back at exactly the same commodity and person from whence the sequence originated, without trading the same commodity twice or "visiting" the same trader more than once. While this may seem a little strange, it begins to express the ability of a market system to combine minimized "shoe leather" with minimum restriction upon permitted trades. Further, it is of paramount mathematical importance for providing the identity of a trading system, and is a prerequisite for the definition of value. Without this condition, a system of traders cannot themselves constitute a stable value index, because in any other case their trading activities will be thwarted by the existence of culs-de-sac that terminate sequences of trade.

IV. Money as the abstract algebra of value

A social theory of value would demonstrate that even with the existence of socially specified commodity identities, and the institution of a connected social trading network, the problem of value is still not resolved. For instance, in Ostroy and Starr (1974), there is still no ability to discuss "price," and the mere introduction of debt undermines all of their results. In a social theory of value, it would be recognized that a distinct algebra of value is required that cannot depend for its identity upon the characteristics of any particular commodity (Mirowski, 1986). Another way of putting this is: why must any particular bilateral trade have any significant implications for any subsequent bilateral trade? How can any traders gauge their overall gain or loss? To put it just a tiny bit more formally, why should it ever be the case that $p_{ab} \times p_{bc} \times p_{cd} \times p_{de} = p_{ae}$? This question is the province of *abstract algebra*.

In a pure barter economy, any arbitrary trade has no significance for any subsequent trade, mainly because there is no stable natural identity or benchmark against which all trades may be compared. What is required is a transpersonal transtemporal index of gain and loss, which also provides the identity element in the algebra of exchange. Money, by its very nature a social institution, renders the price system an algebraic "field" (Durbin, 1985, ch. 5): namely, the set of {all prices} and two operations, addition (with identity zero) and multiplication (with identity unity, the monetary unit). What this guarantees is that (a) prices of different goods may be added and subtracted—a condition that is *not* guaranteed by their own measurement schemes (see section II)—and (b) trade is treated as if it were a reversible process. This ideal of reversibility provides the benchmark of gain and loss. An analogy could be drawn here with thermodynamics, where ideal reversible processes provide the benchmark for real-world irreversible processes (Ellerman, 1984, p. 246).

Hence, in a very narrowly defined sense, in a social theory of value money *is* value; but precisely because it is socially constituted, its invariance is not guaranteed by any "natural" ground, and must be continually maintained by further social institutions, such as the development of double-entry accounting and financial institutions such as banks. (This sets the social theory of value apart from all previous theories of value, which parlay a skepticism over invariance of the monetary unit into an exile of money from the theory of value. See Mirowski, 1989.) The possibility of mutual gain outside of a zero-sum game in a market economy is to be ultimately attributed to the expansion of the monetary unit, especially through debt creation, which gives rise to irreversible trading schemes through time. However, from the vantage point of the participants, the proximate cause of profit is the attribution of property rights to the proliferation of newly created assets and liabilities in their own balance sheets. This stress on the importance of the legal setting of the algebra of double-entry accounting is derived from Ellerman (1986), although it can be traced back to the work of John R. Commons in the 1930s. What was missing from the older institutionalist tradition, however, was a model that expressed how this expansion of value at the individual level is constrained by the social structures at the level of the market system.

V. Market graphs and the value system

In order to develop the ideas broached in the previous section, we shall draw upon the work of Ellerman (1984, 1988) in formalizing what he calls "market graphs." A *directed graph* $\zeta = \{\zeta_0, \zeta_1, t, h\}$ is a set of ζ_0 nodes numbered $1, \ldots, N$, a set ζ_1 of "arcs" or "arrows" numbered

Figure 1 A path from $j = 1$ to $j = n$

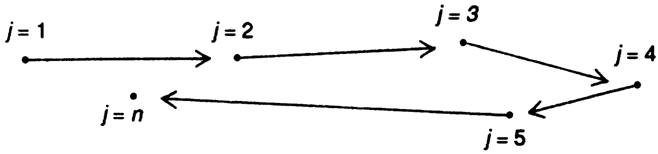
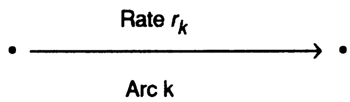


Figure 2 An element of a market graph



$1, \ldots, L$, and head and tail functions which indicate that arc k is directed from its tail node, $t(j)$, to its head node, $h(j)$ (Wilson, 1985; Roberts, 1978; Rockafellar, 1984). It will be assumed below (for the economic reason that virtual self-exchange has no role in a social theory) that no arc is a loop at a node, that is, $h(j) \neq t(j)$. A path from node $j = 1$ to $j = n$ is a sequence of arcs connected at their heads or tails that reach from node $j = 1$ to node $j = n$ (see Figure 1).

A graph is called "connected" if a path exists between any two nodes. The graph in Figure 1 is connected, as one can see by inspection. A closed circular path where no arc occurs more than once is called a "circuit." Figure 1 does not contain a circuit.

In order to endow graphs with economic significance, we shall associate each generic commodity (already indexed $j = 1, \ldots, N$ as above) with a node, and associate each arc with a possible exchange. In order to discuss the price mechanism, we shall associate each arc $k = 1, \ldots, L$ with a nonzero rational number, r_k, called a "rate." Given any arc k, one unit of the commodity at the tail node can be exchanged for r_k units of the commodity at the head node. At this stage, we have no reason to believe that commodities are intrinsically comparable, so that the r_k values should be regarded as merely notional and speculative (who is doing the speculating will be clarified shortly). (See Figure 2.)

It is the object of this section to show that the introduction of money in conjunction with the presence of other specific social arrangements renders this system of "rates" an algebraic group, or, more specifically, a multiplicative group of the nonzero rational numbers. Moreover, it will only be because of money that each rate will be regarded as reversible: that is, if a path traverses an arc in the opposite direction of the arrow, the rate will be treated as the reciprocal $1/r_k$.

Let us define an ideal situation (i.e., in the presence of money) as one where one can define a composite rate over a path as the product of the relevant rates over the path, $r[\alpha] = \Pi_\alpha r_k$. In such a situation, it will be possible to associate a number, p, called a price with each of the nodes/commodities, derived from the set of rates associated with the graph. This triple $\{\zeta, r, p\}$ is what we shall call a "market graph." The important conditions for the existence of such a market graph are that: (1) there are no loops at any node, which means that virtual self-exchange of any commodity is prohibited (this is motivated by the discussion in section II above); (2) the graph ζ must be connected, which means that one can always get to a specific commodity starting at any other (the subject of section III above and some further comments below); and (3) any circuit of exchange that begins and ends at the same commodity results in a composite rate $r[\alpha] = 1$. Under these conditions, Ellerman (1984) proves what he calls the "Cournot–Kirchoff arbitrage theorem" for market graphs:

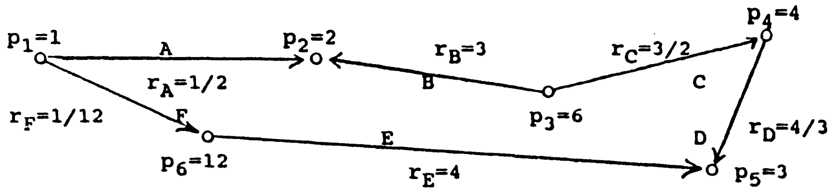
Cournot–Kirchoff arbitrage theorem: Let $\{\zeta, r, p\}$ be a market graph with $r[\alpha]:\zeta_1 \rightarrow T$ taking values in any group T. Then the following conditions are equivalent:

1. There is a price system P derived from the rate system r;
2. The rate system r is path-independent; and
3. The rate system r is arbitrage-free.

This is the fundamental theorem of the social theory of value because it summarizes the conditions under which a market may be considered a *price system* as opposed to a disparate motley of unconnected barter activities. Only when it is always the case that any arbitrary ratio of exchange is consistent with any ratio referring to the same generic commodities but derived from other distinct exchange ratios, can we say that every exchange has calculable implications for every other exchange. This condition is isomorphic to the condition that no arbitrage opportunities are present due to mismatches in the system of prices. Another way of stating this condition is to assert that *value is conserved in the exchange process.*[2]

[2] In Mirowski (1989) I asserted that such conservation principles are central to the understanding of the history of Western economic thought. This link to the Cournot–Kirchoff theorem could be considered one more installment in that argument.

Figure 3 An arbitrage-free market graph



The relationship between the Cournot–Kirchoff theorem and the price system may be rendered more transparent by linking the graph formalism to its matrix representation.[3] Let us define a node–arc incidence matrix S of the graph $\{\zeta_0, \zeta_1, r\}$ as follows:

$$S_{jk} = \begin{cases} r_k & \text{if arc } k \rightarrow \text{node } j; \\ -1 & \text{if arc } k \leftarrow \text{node } j; \\ 0 & \text{otherwise.} \end{cases}$$

The matrix S has dimensions $N \times L$, with the rows representing the nodes/commodities, and the columns representing the arcs/exchange possibilities. If the matrix is descriptive of a market graph meeting all the conditions for the Cournot–Kirchoff theorem, then the rank of S will be equal to $N - 1$. (This is a quick procedure to check for the arbitrage-free condition.) Moreover, let us define a price vector $p = \{p_1, \ldots, p_N\}$ where price p_j is associated with the commodity node indexed j. Then the arbitrage-free system of prices can be solved as follows:

$$pS = p_{j_0} \cdot r_k - p_{j_1} = 0.$$

Of course, the zero price vector is always a trivial solution to this problem, but it should be rejected as the case where a relative price system does not exist. If the product of the price vector and the incidence matrix is nonzero, then those amounts represent the "arbitrage profit" in money terms from making a circuit of exchanges beginning and ending at that particular node.

3 Another advantage of the incidence matrix approach is that it allows the incorporation of certain results from parts of advanced Sraffian analysis without adopting the conventional interpretations of the "standard commodity," etc. (Krause, 1982). For instance, one can show that the Sraffian requirement that the input/output matrix be indecomposable is primarily dictated by the fact that a fully connected incidence matrix is the primary prerequisite of a successful trading system.

Perhaps a numerical example will help demonstrate the intimate relationship between arbitrage theory, directed graphs, and the linear algebra. In Figure 3 we have the graph-theoretic representation of a very simple six-commodity market. The nodes are different commodities numbered from one to six, and labeled with their attached prices. Each arc is labeled with a letter from A to F and with its attendant rate of exchange between the relevant nodes. From inspection, one can see the entire graph consists of a single circuit; also by inspection one can tell it is arbitrage-free since the composition of rates around the entire circuit multiplies to unit, that is:

$$r[\alpha] = (1/2)\,(1/3)\,(3/2)\,(4/3)\,(1/4)\,(12) = 1.$$

The entire structure of the graph can be expressed in its corresponding node–arc incidence matrix:

$$\begin{array}{cc} & \begin{array}{cccccc} A & B & C & D & E & F \end{array} \Leftarrow k \\ \begin{array}{c} 1 \\ 2 \\ 3 \\ 4 \\ 5 \\ 6 \end{array} & \begin{bmatrix} -1 & 0 & 0 & 0 & 0 & -1 \\ 1/2 & 3 & 0 & 0 & 0 & 0 \\ 0 & -1 & -1 & 0 & 0 & 0 \\ 0 & 0 & 3/2 & -1 & 0 & 0 \\ 0 & 0 & 0 & 4/3 & 4 & 0 \\ 0 & 0 & 0 & 0 & -1 & 1/12 \end{bmatrix} \\ \Uparrow & \\ j & \end{array}$$

Solving for $pS = 0$, we arrive at the price vector,

$$p = [1, 2, 6, 4, 3, 12],$$

which corresponds to the prices attached to each of the nodes.

What sort of economy is described by this formalism? Prices are adjusted through system bids and offers in order to attempt to achieve a system of reversible, arbitrage-free exchange with a positive price vector $\mathbf{p^*}$, which can be found by solving $\mathbf{p^*S} = 0$. Traders do not engage in this activity out of benevolence, but rather in self-interested (but *not* maximizing!) pursuit of arbitrage profits.

Far from being a simple mechanical calculation, the persistent attempt to reconcile money prices with ratios of commodity quantities and to attain the "arbitrage-free" state is the very core of an institutionalist value theory. This model describes a system that is both "intentional," in the sense that numerous social institutions from the standardized weights and measures to the monetary authority all strive to structure the value unit, and also "unintentional," in that the mathematical structure of the value system falls out of the persistent individual attempts to make arbitrage gains in trade. The arbitrage-free state thus arrived at embodies what classical economists tried to express as the conservation of value in exchange. These conservation principles are built in at the very foundations of double-entry accounting practices; hence, the approach of actual exchange relationships to the accounting system is the very definition of a stable unit of value.

VI. Against equilibrium

Given the almost neurotic concern of orthodox mathematical economics with the physical metaphor of "equilibrium," it may be appropriate here to insist that in no case should the vector **p*** be regarded as an equilibrium price vector in this theory.[4] While absence of further arbitrage opportunities would signal a fully path-independent system within its own terms, real-time approach to that state would inevitably alter the underlying r_k values, and consequently the resulting **p***. The "law of one price" has never made much sense in a dynamic context in orthodox theory, and much the same situation holds here (Bausor, 1986).

It may help to recall who is responsible for the r_k values, which provide the raw material for the Cournot–Kirchoff theorem. From one potential point of view, the r_k is the only psychological term in the entire theory, representing the conjectures of individuals with regards to the exchange ratios they might achieve within the limited ambit of their "social traffic." If this indeed were so, then the full social theory of value would require concatenation of the social traffic matrix with the node–arc incidence matrix, along the lines of the matrix Ω:

$$\Omega = \begin{bmatrix} \bar{z}_1 & S^1 \\ \bar{z}_2 & S^2 \\ \cdot & \\ \cdot & \\ \cdot & \\ \bar{z}_m & S^m \end{bmatrix}$$

[4] Or to address those of mathematical bent directly: there is no sense in trying to turn $pS = 0$ into some sort of difference or differential equation, and then applying Liapunov techniques, nonlinear dynamics, or any other faddish device taken over from physics to it. On this related issue, see Mirowski, 1990b.

where $\bar{z}_i$ represents the ith row of the matrix **z**, distributed along the diagonal of a square matrix of order $N \times N$ with zeroes on the off-diagonal, and S^m is the node–arc incidence matrix for each individual, indexed $1, \ldots, m$. One could then apply the Cournot–Kirchoff theorem to the matrix Ω, but it should already be apparent that there is little reason to think that the prior conditions of connectivity would hold, given our discussion in section III. It would be prudent, rather, to conclude that one cannot depend upon some mechanical adjustment procedure applied to individual psychologies to succeed in explaining market coordination.

An alternative interpretation of the r_k values is to regard them as the conjectures of various trade coordinators (as identified by their locations in the **z** matrix) in their relationships with one another. These coordinators would be responsible for enforcing the invariance of price in their locality with regard to the larger public, as well as for forging the connectivity of the market graph (hence the prevalence of "sticky" price behavior among such units). But even in this instance we should not expect to find true invariance. The intrinsic fluidity of the system militates against any strict causal directionality from the r_k values to the reigning prices.

Consequently, the "arbitrage-free" state is never realized in a capitalist system. Its existence would demand a stasis not encouraged within the system, and in any event would imply that globally trade was a zero-sum game, and therefore would rule out the existence of profit generated at the system level. Instead, the various macro determinants of system expansion discussed in section IV persistently perturb trade away from any approximation to the arbitrage-free state and feed back upon the definition of the subsequent state. Thus, prices never "approach" any fixed or deterministic point, and time series of prices will inevitably appear stochastic from the perspective of both the actors and the economic analyst. But does this then imply that the system wanders aimlessly? Why have we bothered with the whole formalism of market graphs, social traffic matrices, and all the rest?

These questions bring us to the "postmodern" character of the social theory of value. Economists seem to think that "models" exist to capture the reality of the situation in which the economic actor finds himself embedded. This unobtrusive postulate of a fixed reality, independent of the interpretative engagement of human beings, has undoubtedly been fostered by the persistent tendency of economists to imitate physicists and their models. The role of our mathematics is different: it is intended to describe the interpretative structures of modern market institutions without implying that these structures are permanent, inevitable, or deterministic. This "model" exists to give an impression of what it is like to "read" the economy without presuming there is a single fixed text to be read, or a single correct interpretation at which we can arrive.

Hence, the function of the mathematics of market graphs is not to describe the prices that actually exist, but to explain why prices are constituted as a field over the rational numbers. The numerical character of prices is an artifact of the way our society has evolved for "reading" the consequences of our actions in the economic sphere: so I have opted to purchase this stereo system? The price system then provides a way for me to "read" the consequences of this action for any other actions I may take. The consequences range from the rather banal (with my salary, I am not able to buy a computer this month, or even next month) to some rather subtle cues. They include:

a. The effect of my purchase upon the identity of the commodity is disregarded in the process of exchange.
b. Buying nothing is treated as costing me nothing.
c. Prices over time and space can be added, but the order in which the items are presented for purchase are not treated as having an impact upon the sum total paid for them.
d. If it is possible to "return" the item, then the net result should be zero.

The assertion is not that any of these principles are "true"; indeed, for most people in most instances, they are not. The point is rather that in order to read the consequences of our actions, some forms of change have to be ignored, or bracketed, or exiled to the margins of the text. Something must be treated as effectively invariant, even as we know all along it is not. In this sense, the price system should be situated upon the same epistemic level as double-entry bookkeeping (if indeed they are not merely two sides of the same historical phenomenon); the mathematics merely makes this more transparent.

VII. The fundamentally stochastic character of the economy

How is it possible that market actors live a schizophrenic existence of treating value as if it were invariant, and yet are also fully cognizant that they cannot depend upon the invariants? This question leads us into the realm of probability theory and its intimate relationship with the social theory of value.

The biggest source of disruption to value invariants is also the primary economic motivation in a market system: the pursuit of profit. At the aggregate level, there is a trade-off between the expansion of value through debt creation and the breakdown of the value invariant through inflation; the relationship is *socially constructed* and therefore non-mechanical, and thus further social institutions are required to intervene continually to offset one or the other trend. The overriding social problem of all market-oriented societies is to find some means to maintain the working fiction of a monetary invariant through time, so that debt contracts (the ultimate locus of value creation, as indicated above) may be written in terms of the unit at different dates.

This can be thought of as a "homeostatic system": fluctuations in the underlying value unit are unavoidable in a system of private contracts, both because anyone can charge a "time premium," and because the arbitrage-free state is never attained, so everyone is calculating with faulty prices. Nevertheless, if debt creation is kept within certain limits, the system functions as if a value invariant exists. It is important to note that, in this system, any time series of specific prices is inherently and irreducibly stochastic: unlike in neoclassical "efficient markets theory," the fluctuations are not due to external "noise" superimposed upon "fundamentals," but to the internal operation of the value system. Such an alternative metaphor for what the market does may be more attractive in a period that acknowledges that the earlier reconciliation of the image of equilibrium with stochastic ideas has failed (Leroy, 1989).

It is possible to link the social theory of value to the findings of Benoit Mandelbrot (1963a, 1963b) that price distributions do not generally conform to Gaussian stories of the separation of "noise" from signal (Mirowski, 1990b). The task of a social theory of value would then be to reconcile "arbitrage theory"—that is, the conviction that prices can be brought closer to their underlying determinants or coherency conditions—with the knowledge that the underlying determinants are not given by nature, but are constituted by the very process of exchange itself, and therefore are intrinsically stochastic.

The crowning achievement of a social theory of value would be to demonstrate that the very axioms of probability are themselves not innate, but rather are jointly constructed with the value system as part of the operation of markets described above. Following a hint in Ellerman (1984, p. 254), and building from the insights of de Finetti (1974), we can show that the conventional Kolmogorov axioms of probability theory (namely, $0 \leq P(x) \leq 1$; $P(\Omega) = 1$; and $P(x \cup y = P(x) + P(y))$ can be derived from the simple requirement of "coherence" in betting: that is, the "arbitrage-free" condition extended to all monetary wagers. Here "risk" is a socially constructed concept (Douglas, 1985) in the sense that the existence of an arbitrage-free value unit plus the requirement that actors not allow "Dutch books" to be made persistently against them teaches market participants to behave in the aggregate as if the Kolmogorov axioms described the world of uncertainty that they face. Again, this condition is only approached, and never attained, which is why so many neoclassical researchers have "discovered" that individuals in control situations do not "obey" the laws of probability in a von Neumann–Morgenstern framework. In an institutionalist theory of value, probability theory is relegated to the same epistemic level as double-entry bookkeeping: while not "true," it is the pragmatic instrumentality by which market transactors reckon their gains and losses. Thus, the laws of probability are intimately bound up with a money economy, not because people's neurons come hard-wired for the Kolmogorov axioms, but rather because the axioms are simply one more projection of the arbitrage-free conditions in a social theory of value.

REFERENCES

Anderson, Benjamin. *Social Value*. New York: Kelley, 1966 [1911].

Bausor, Randall. "Time and Equilibrium." In *The Reconstruction of Economic Theory*, P. Mirowski, ed. Boston: Kluwer, 1986.

Bergé, Claude, and Ghouli-Houri, A. *Programming, Games and Transportation Networks*. New York: Wiley, 1962.

Clower, Robert. *Money and Markets*. New York: Cambridge University Press, 1984.

Davidson, Paul. *Money and the Real World*, 2d ed. New York: Wiley, 1978.

Debreu, Gerard. "Economic Theory in a Mathematical Mode." *American Economic Review*, 1984, *74*, 267–278.

de Finetti, Bruno. *Theory of Probability*. New York: Wiley, 1974.

de Jong, Frits. *Dimensional Analysis for Economists*. Amsterdam: North Holland, 1967.

Douglas, Mary. *Risk Acceptability According to the Social Sciences*. New York: Russell Sage, 1985.

———. *How Institutions Think*. Syracuse, NY: Syracuse University Press, 1986.

Durbin, John. *Modern Algebra*, 2d ed. New York: Wiley, 1985.

Eckalbar, John. "Money, Barter and Convergence to Competitive Allocation." *Journal of Economic Theory*, 1984, *32*, 201–211.

Ellerman, David. "Arbitrage Theory: A Mathematical Introduction." *SIAM Review*, 1984, *26*, 241–261.

———. "Appropriation and Property Theory." In *The Reconstruction of Economic Theory*, P. Mirowski, ed. Boston: Kluwer, 1986.

———. "An Arbitrage Interpretation of Classical Optimization." Unpublished manuscript, Tufts University, 1988.

Grossman, I., and Magnus, W. *Groups and Their Graphs*. New York: Mathematics Association of America, 1964.

Heilig, Klaus. "The Dutch Book Argument Reconsidered." *British Journal for the Philosophy of Science*, 1978, *29*, 325–346.

Krause, Ulrich. *Money and Abstract Labour*. London: NLB Verso, 1982.

Kula, Witold. *Measures and Men*. Princeton, NJ: Princeton University Press, 1986.

Leroy, Stephen. "Efficient Capital Markets and Martingales." *Journal of Economic Literature*, 1989, *27*, 1583–1621.

Luce, H., and Narens, L. "Measurement Theory." In *The New Palgrave*, ed. J. Eatwell, et al. London: Macmillan, 1987.

Mandelbrot, Benoit. "New Methods in Statistical Economics." *Journal of Political Economy*, 1963a, *71*, 421–440.

———. "The Variation of Certain Speculative Prices." *Journal of Business*, 1963b, *36*, 394–419.

Mirowski, Philip. "Mathematical Formalism and Economic Explanation." In *The Reconstruction of Economic Theory*, P. Mirowski, ed. Boston: Kluwer, 1986.

———. *Against Mechanism*. Totawa, NJ: Rowman and Littlefield, 1988.

———. *More Heat Than Light: Economics as Social Physics, Physics as Nature's Economics*. New York: Cambridge University Press, 1989.

———. "Learning the Value of a Dollar: Conservation Principles and the Social Theory of Value," *Social Research*, 1990a, *57*, 687–717.

———. "From Mandelbrot to Chaos in Economic Theory." *Southern Economic Journal*, 1990b, pp. 289–307.

———. "The How, the When and the Why of Mathematics in Economics." *Journal of Economic Perspectives*, 1991, forthcoming.

Osborne, D. "On Dimensional Invariance." *Quantity and Quality*, 1978, *12*, 75–89.

Ostroy, J. "Money and General Equilibrium Theory." In *The New Palgrave*, ed. J. Eatwell, et al. London: Macmillan, 1987.

Ostroy, J., and Starr, R. "Money and the Decentralization of Exchange." *Econometrica*, 1974, *42*, 1093–1113.

Roberts, Fred. *Graph Theory and Applications to Problems of Society*. Philadelphia: SIAM, 1978.

———. *Measurement Theory*. Reading, MA: Addison-Wesley, 1979.

Rockafellar, R. T. *Network Flows and Monotropic Optimization*. New York: Wiley, 1984.

Rosen, Joe. "Fundamental Manifestations of Symmetry in Physics." *Foundations of Physics*, 1990, *20*, 283–307.

Schick, Frederick. "Dutch Bookies and Money Pumps." *Journal of Philosophy*, 1986, *83*, 112–119.

Swamy, M., and Thulasiraman, K. *Graphs, Networks and Algorithms*. New York: Wiley, 1981.

Wilson, Robin. *An Introduction to Graph Theory*, 3d ed. Essex: Longman, 1985.